Mohamed Amine Chaieb

Computer Engineering Student: Software & AI

3rd-year Computer Engineering student at ENSI • Seeking a PFE (end-of-studies) internship, flexible dates

m.amine.chaieb@gmail.com • +216 29 663 667
linkedin.com/in/mohamedaminechaieb • github.com/mohammedaminechaieb
amine-portfolio-cmtt.onrender.com

PROFILE

Computer engineering student with a broad profile: software engineering (algorithms, clean architecture, tests, 87% coverage on critical backend modules of a team project), applied AI (LLM agents, speech recognition, intent classification, OpenCV) and embedded/IoT (ESP32). Projects in Python, C++, JavaScript, Dart and Kotlin.

SKILLS

Languages: Python, JavaScript, C/C++, Dart, SQL
Frontend & mobile: React, Node.js/Express, FastAPI, Flutter
Practices: unit/integration testing, state machines, REST design, Git
AI & embedded: faster-Whisper, Piper, LLM APIs (Mistral via OpenRouter, Gemini), intent classification, OpenCV, ESP32, n8n
Familiar with: TypeScript, Angular, Kotlin, Docker

EXPERIENCE

Mobile Developer Intern: ID Card Detection & Autocapture Summer 2026 (6 weeks)
Codingaholic (Avatar-ID app)

- Designed and built, in 6 weeks, a Flutter module for the Avatar-ID identity app that detects a Tunisian ID card in the live camera feed and captures both sides automatically (OpenCV through opencv_dart).
- Clean Architecture (feature-first) with 177 passing automated test cases across four test tiers.

Capstone Design & Development Project (PCD): GOB 2025 – 2026
ENSI

- Co-built GOB in a team of 3 (capstone, 2025-2026): a multilingual (FR/EN/AR) AI voice agent that turns spoken commands into real actions (email, reminders, an ESP32 robot) through n8n workflows.
- Safe actions: finite state machine with user confirmation and HMAC-SHA256 signatures before execution; the n8n workflows passed 116 of 120 tests.
- Tested with pytest: 20 end-to-end scenarios and 87% coverage on critical backend modules; conversation latency about 3.8 s (team results).

Research Assistant: Verbal Aggression Detection in Smart Cities Jun–Jul 2025 (6 weeks)
RIADI Laboratory, ENSI

- Designed an end-to-end architecture: ESP32 with a microphone amplifier, REST API, MongoDB, CNN, LSTM and CRNN acoustic models, speech-to-text language analysis and a fusion layer.

PROJECTS

- **Android suite** (core logic done, polishing and testing in progress): AuraLauncher with HaptiKit, NightDeck, StatusBar+, SoundSkin and LockForge, bundled around a custom launcher (Kotlin, Flutter, Jetpack Compose).
- **AmineOS** (portfolio): interactive desk-and-desktop portfolio built with Next.js and TypeScript, deployed on Render; earlier version in React, Node.js/Express and MongoDB Atlas.
- **The System** (in testing): habit, sleep and fitness tracker with dashboard, streaks, a skippable soft-lock screen and an AI dashboard powered by the Gemini API.
- **CTF challenges**: designed advanced challenges for TuniHack 10.0 and ENSI club events; regular CTF player on TryHackMe and HackTheBox.

EDUCATION

National Engineering Degree, Computer Engineering • ENSI, University of Manouba Sept 2024 – 2027 (expected)
Preparatory Cycle, Mathematics–Physics • Faculty of Science Tunis, University El Manar 2022 – 2024

COMPETITIONS & ACTIVITIES

- **Go Data Science 4.0** hackathon (IEEE CIS ENSI): ranked 37th (Feb 2025).
- Member of the ENSI Competitive Programming Club and the Open Source Software ENSI Club; participant, ENSI Programming Contest 4.0 (Dec 2024).
- **Organizer**: TuniHack 10.0 (sponsorship, logistics, CTF challenges; Jan 2025) and Code & Conquer 2nd ed. (sponsorship team; Apr 2025).

CERTIFICATIONS & LANGUAGES

Programming Hero: Python (Oct 2024) • Programming Hub: Ethical Hacking (Oct 2024)
Arabic (native) • French (B2) • English (C1)